Program : **Diploma inPolymer Technology**	
Course Code :**3095**	Course Title: **Basic Mechanical Engineering**
Semester : **3**	Credits: **No Credit**
Course Category: **Program Core**	
Periods per week: **4 (L:3,T:1,P:0)**	Periods per semester:**60**

Course Objectives:

- To perceive classification metals, alloys and fuels.
- To recognize important properties of steam and components of steam boilers.
- To assimilate working of air compressors and refrigeration systems.
- To realize power transmission devices and lubrication.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basics of thermodynamics Mechanics of solids		Applied Physics I	I
Metallic bonds Calorific value Lubricants		Applied Chemistry	I

Course Outcomes:

On completion of course, the student will be able to:

COn	Description	Duration (hours)	Cognitive Level
CO1	Differentiate types of metals & alloys and classify various fuels	15	Understanding
CO2	Explain various properties of steam and working of steam boilers and their components	14	Understanding
CO3	Describe working of air compressors and refrigeration systems	14	Understanding
CO4	Identify uses of power transmission devices and discuss lubricants & lubrication systems	15	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2			2				
CO3			2				
CO4			2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

<table>
<tr><th>Module Outcomes</th><th>Description</th><th>Duration (Hours)</th><th>Cognitive Level</th></tr>
<tr><td>CO1</td><td colspan="3">Differentiate types of metals & alloys and classify various fuels</td></tr>
<tr><td>M1.01</td><td>Illustrate classification of metals and their properties</td><td>4</td><td>Understanding</td></tr>
<tr><td>M1.02</td><td>Realize alloys and alloy steels</td><td>5</td><td>Understanding</td></tr>
<tr><td>M1.03</td><td>Describe classification of fuels and their requirements</td><td>2</td><td>Understanding</td></tr>
<tr><td>M1.04</td><td>Assimilate calorific value of fuels</td><td>4</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Classification of metals – mechanical properties of metals - types of cast iron and steels.
Need for alloying steel with other elements – list properties of alloy steels - need for the use of non-ferrous metals and alloys in engineering operations - suitability of different metals and alloys for making polymer processing machine parts and moulds.
Classification of fuels - requirements of a good fuel - merits and demerits of solid, liquid and gaseous fuels.
Explain higher and lower calorific value of fuels (simple problems).</td></tr>
<tr><td>CO2</td><td colspan="3">Explain various properties of steam and working of steam boilers and their components</td></tr>
<tr><td>M2.01</td><td>Comprehend steam and its properties</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Describe classification of boilers</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Summarize boiler mountings and accessories</td><td>5</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Outline the package boilers and boiler drafts</td><td>2</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test - I</td><td>1</td><td></td></tr>
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CO3	**Describe working of air compressors and refrigeration systems**		
M3.01	Describe functions of air compressor, their classification and uses of compressed air	3	Understanding
M3.02	Illustrate reciprocating compressors	4	Understanding
M3.03	Discuss various types of rotary compressors	3	Understanding
M3.04	Outline principle and working of refrigeration system	4	Understanding

Contents:

Function of an air compressor - uses of compressed air - classification air compressors.

Working of reciprocating compressors with simple sketches (single stage and multistage) - advantages of inter-cooling.

Differentiate positive and non-positive displacement rotary compressors - various types of rotary compressors-Centrifugal, Axial, Vane, Screw compressors and Roots blower (simple sketches).

Working of vapour compression system (block diagram) – important components of vapour compression refrigeration system and their functions - Define the unit of refrigeration and COP - List various types of refrigerants - Justify the selection of refrigerants for different systems - industrial application of refrigeration system in polymer industries.

CO4	**Identify uses of power transmission devices and discuss lubricants & lubrication systems**		
M4.01	Describe belt and chain drives	5	Understanding
M4.02	Outline gear drive and classification of gears	4	Understanding
M4.03	Discuss various types of bearings	4	Understanding
M4.03	Summarize lubricants and lubrication systems	2	Understanding
	Series Test - II	1	

Contents:

Application of belt drives - classification of belts (simple sketches) - various types of belt drives (simple sketches) - define velocity ration of belt drive - compare V-belt and Flat belt drives - application of chain and sprocket drive - types of chain drives.

Concept of friction wheel - advantages and disadvantages of gear drive – classification of gear wheels and their function-spur, helical, herringbone, bevel and worm gears(simple sketches) - define gear train and list different types.

Bearings - classification - bearings according to the direction load such as journal, foot-stop, thrust and collar bearings (simple sketches) – construction(simple sketches) and advantages of the brush, ball, roller and needle bearings.

Explain hydraulic and pneumatic power transmission.

Function of lubricants - desired properties of lubricants - types of lubricants - different types of engine lubricating systems-splash and pressure lubrication systems.

Text /Reference:

T/R	BookTitle/Author
T1	Basic Mechanical Engineering - J Benjamin
R1	A Textbook of Workshop Technology:Manufacturing Processes - R.S Khurmi& JK Gupta
R2	Elements of Heat Engines Vol. I - R.C.Patel& C.J. Karamchandani
R3	Refrigeration & Air conditioning - P.L Ballaney
R4	Theory of Machines - S.S Rattan

Online resources:

Sl. No	Website Link
1	www.researchgate.net
2	www.scribd.com
3	https://ocw.mit.edu/
4	https://ocw.mit.edu
5	https://learnmech.com